

Basic Information

Experiment No.: 07

Date: 08-10-2026

Title: Java HashMap and Its Methods

Objective / Aim

To implement "HashMap" in Java and demonstrate operations such as adding, accessing, removing, searching, retrieving keys and values, checking size, and clearing the map.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

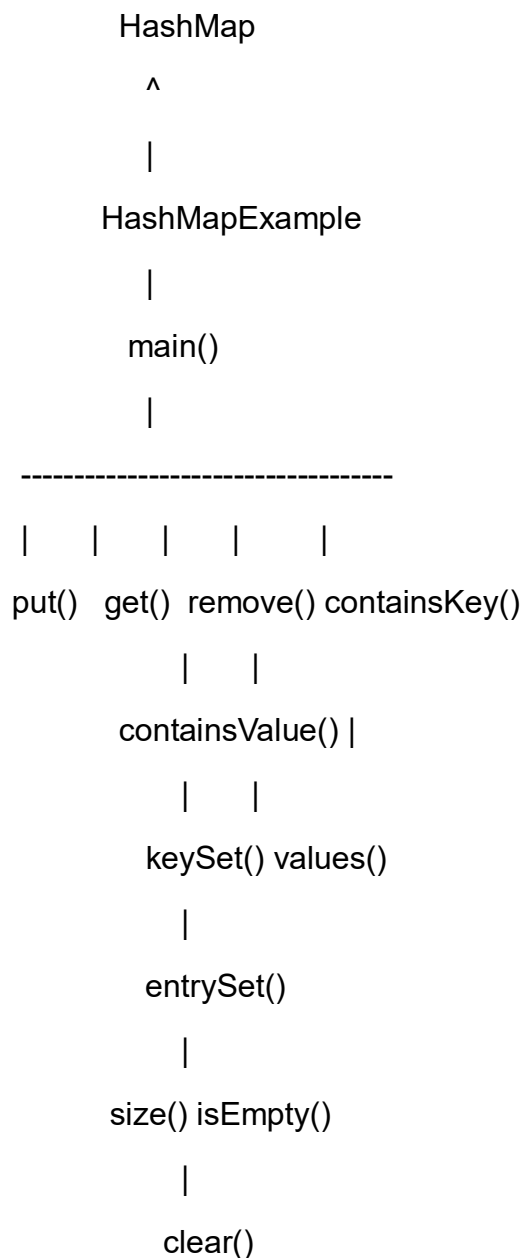
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "HashMap" with integer keys and string values.
2. Add key-value pairs using "put()".
3. Access values using "get()" and "getOrDefault()".
4. Check keys and values using "containsKey()" and "containsValue()".

5. Remove a mapping using "remove()".
 6. Display keys using "keySet()".
 7. Display values using "values()".
 8. Display key-value pairs using "entrySet()".
 9. Check the map size and empty status.
 10. Remove all mappings using "clear()".
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UML Class Diagram



|
getOrDefault()

Source Code

```
package collectionframework;

import java.util.HashMap;

public class HashMapExample {
    public static void main(String[] args) {

        HashMap<Integer, String> map = new HashMap<>();

        // put()
        map.put(101, "Rahul");
        map.put(102, "Priya");
        map.put(103, "John");

        System.out.println("HashMap: " + map);

        // get()
        System.out.println("Value of key 102: " +
            map.get(102));

        // getOrDefault()
        System.out.println("Value of key 104: " +
            map.getOrDefault(104, "Not Found"));
```

```
// containsKey()
System.out.println("Contains key 101: " +
    map.containsKey(101));

// containsValue()
System.out.println("Contains value John: " +
    map.containsValue("John"));

// remove()
map.remove(103);
System.out.println("After remove: " + map);

// keySet()
System.out.println("Keys: " + map.keySet());

// values()
System.out.println("Values: " + map.values());

// entrySet()
System.out.println("Entries: " + map.entrySet());

// size()
System.out.println("Size: " + map.size());

// isEmpty()
System.out.println("Is Empty: " + map.isEmpty());

// clear()
map.clear();
```

```
        System.out.println("After clear: " + map);  
    }  
}
```

Compilation & Execution Commands

Save the file as:

HashMapExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

```
javac HashMapExample.java
```

Run:

```
java collectionframework.HashMapExample
```

Output Screen

Note: "HashMap" does not guarantee insertion order, so the order may vary.

HashMap: {101=Rahul, 102=Priya, 103=John}

Value of key 102: Priya

Value of key 104: Not Found

Contains key 101: true

Contains value John: true

After remove: {101=Rahul, 102=Priya}

Keys: [101, 102]

Values: [Rahul, Priya]

Entries: [101=Rahul, 102=Priya]

Size: 2

Is Empty: false

After clear: {}

Result

The "HashMap" class was successfully implemented, and operations such as "put()", "get()", "getOrDefault()", "remove()", "containsKey()", "containsValue()", "keySet()", "values()", "entrySet()", "size()", "isEmpty()", and "clear()" were successfully demonstrated.